Nominal Anchors and Real Consequences: Inflation Targeting, Debt, and Investment

preliminary

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What we do

secular trends in macro-finance since the 1980s often explained by real factors (demography, productivity, capital risk)

we ask: can the *monetary policy regime* itself play a role?

develop GE connections between inflation targeting and wider historical trends:

1. decline in long-term real interest rates on nominal bonds
2. increase in private (nominal) debt relative to GDP
3. investment 'drought' (declining I/Y)
4. stability of risky real returns to capital
5. strengthening of inflation targeting since the 1980s ($\phi_\pi \uparrow$)

can the monetary policy regime (5) be a unified explanation of (1)–(4) in a GE setting?

Preview of the mechanism

credible inflation targeting $\implies$ lower inflation volatility

- nominal bonds become *safer* in real terms
- $\implies$ inflation risk premium + interest rate risk declines $\implies$ $R_0^* \downarrow$
- $\implies$ savers supply more savings in nominal assets $\implies$ debt $\uparrow$
- $\implies$ capital becomes relatively *more* risky $\implies$ $K \downarrow$, $\mathbb{E}_0[R_0^K] \uparrow$

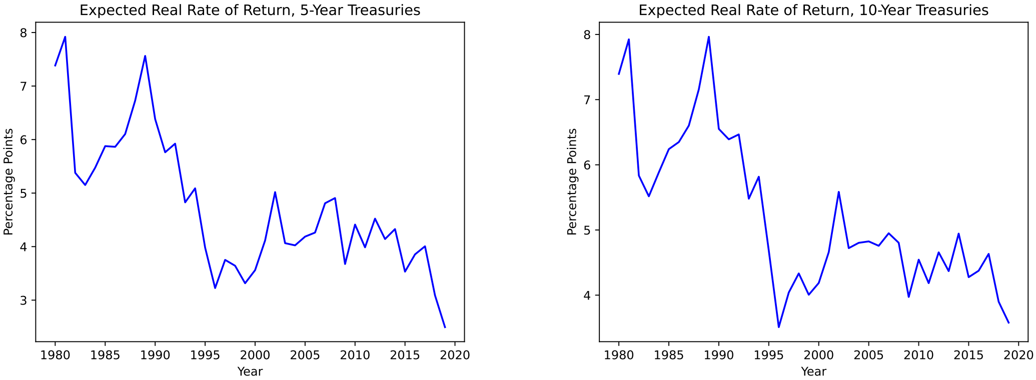
successful nominal anchoring transforms the *risk characteristics* of nominal assets, triggering portfolio reallocation away from risky capital toward safer nominal bonds

Roadmap

1. Stylized facts since the 1980s
2. A three-period macro-finance model
3. Analytical results: zero debt stochastic steady state
4. Analytical results: positive (nominal) debt
5. Conclusion

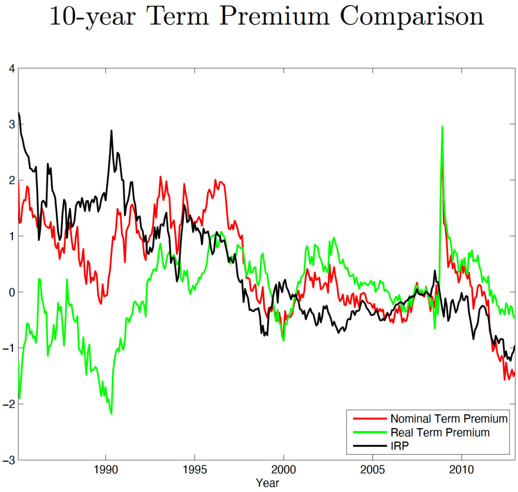
Fact 1: Decline in long-term rates

expected real yields on US nominal bonds



Fact 1: Decline in long-term rates

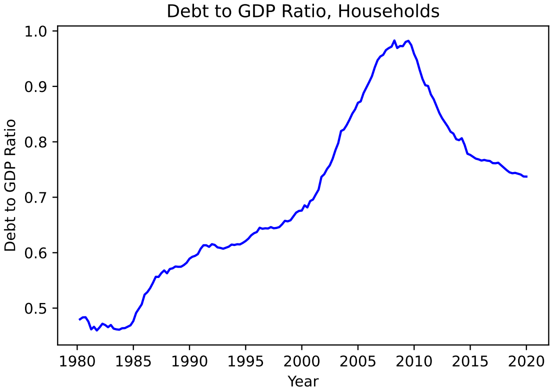
Inflation Risk Premium for the UK



Abrahams, Adrian, Crump, and Moench (2016) for the UK.

Fact 2: Rise in private (nominal) debt

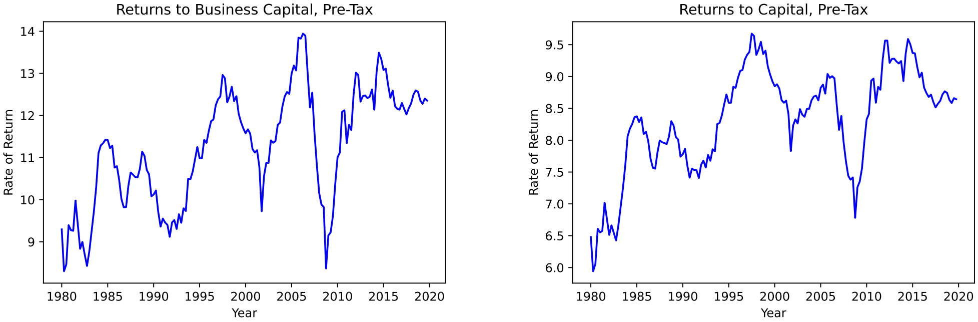
household debt relative to GDP in the US increased since 1980



Source: Our calculations based on Mian Straub & Sufi (2023)

Fact 3: Returns to capital stable or rising

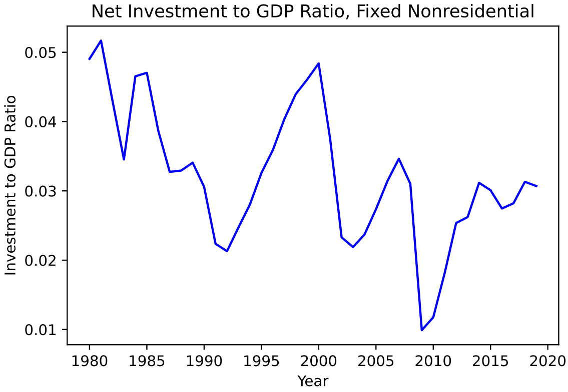
real return to capital, US data



Source: based on Gomme, Ravikumar, & Rupert (2019)

Fact 4: Investment drought

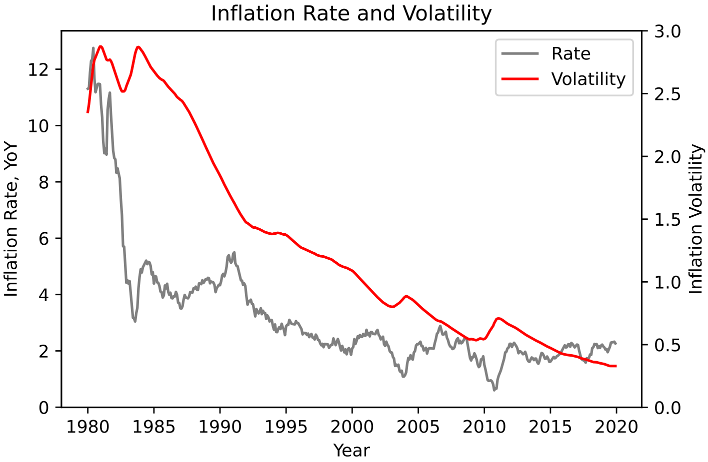
investment to GDP declined



Source: our calculations

Fact 5: Strengthening of inflation targeting

post Volcker, central banks adopt inflation targeting. US data (π, level and rolling sd)



Model Overview

Three Periods: $(t = 0, 1, 2)$, flexible price economy

Agents

- Borrower and Savers
- Central bank (Taylor rule)

Assets

- Short-term nominal bond (zero supply)
- Long-term nominal bond
- Capital: illiquid, cannot sell at date 1

Endowments

- exogenous endowment at all dates

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Shocks

All uncertainty resolves at date 1

Aggregate Risk

- Preference shock ξ_1
- Source of inflation volatility
- inflation risk premium

Idiosyncratic Liquidity Risk on Savers

- Prob. λ: forced to liquidate at date 1
- Low income y_L in liquidity state
- interest rate risk on long-term bond

Environment: Agents and Preferences

Two types $j \in \{b, s\}$: borrowers and savers, each mass one

- Borrowers: discount factor β_b^t, with β_b small (always constrained)
- Savers: discount factor $\beta^t \xi_t$, where ξ_t is a stochastic preference shock

Preference shock ξ_1:

- $\xi_0 = 1$, $\xi_1 \sim [\xi_L, \xi_H]$ with $\mathbb{E}[\xi_1] = 1$, $\mathrm{Var}(\xi_1) > 0$
- Date-2 persistence: $\xi_2 = \xi_1^\rho$ with $\rho \in (0, 1)$
- All uncertainty resolves at date 1

Idiosyncratic Liquidity Shock

Each saver draws $\eta \in \{0, 1\}$ (wt on date 2 utility) at beginning of date 1, i.i.d., independent of ξ_1:

	Impatient ($\eta = 0$)	Patient ($\eta = 1$)
Probability	λ	$1 - \lambda$
Date-1 endowment	y_L	$y_H \geq y_L$
Long bonds at date 1	Sells at Q_1^L	Holds to maturity

Saver's date-0 utility:

$$U_s = u(C_{s,0}) + \beta\,\mathbb{E}\Big[\xi_1\Big(\lambda\, u(C^I_{s,1}) + (1-\lambda)\,\big[u(C^P_{s,1}) + \beta\,\xi_1^{\rho-1}\,u(C^P_{s,2})\big]\Big)\Big]$$

$u(C) = C^{1-\gamma}/(1-\gamma)$

Assets

1. **Short-term nominal bond** — issued at date 0, pays $(1 + i_0^S)$ at date 1
    - Zero supply: pins short-term real interest rate
    - Real return: $(1 + i_0^S)/\Pi_1$ — exposed to date-1 inflation
2. **Long-term nominal bond** — issued at date 0 at price Q_0^L, pays 1 at date 2
    - Liquid: can be resold at date 1 at price $Q_1^L = \frac{1}{1+i_1^S}$
    - Impatient type sells at $Q_1^L \Rightarrow$ exposed to both inflation risk and interest rate risk
3. **Real capital** K — invested at date 0, produces $\bar{A}K^\alpha$ at date 2
    - Illiquid: cannot be sold or pledged at date 1
    - No inflation exposure

Saver's Budget Constraints

Date 0 (before learning η):

$$P_0 C_{s,0} + P_0 K + Q_0^L b^L = P_0 Y_0$$

Date 1, Impatient type (λ): sells all long bonds, consumes everything

$$P_1 C_{s,1}^I = P_1 y_L + Q_1^L b^L$$

Date 1, Patient type $(1-\lambda)$: receives y_H, holds bonds to maturity, saves s_1

$$P_1 C_{s,1}^P + \frac{\lambda}{1-\lambda} Q_1^L b^L = P_1 y_H$$

$$P_2 C_{s,2}^P = P_2 \bar{A} K^\alpha + P_2 Y_2 + \frac{b^L}{(1-\lambda)}$$

Borrowers

Issue long-term nominal bonds, face debt limit

$$b^L/P_0 \leq d_0/R_0^{L*}$$

- Consume at dates 0 and 2 only:

$$P_0 C_{b,0} = P_0 Y_{b,0} + Q_0^L b^L, \quad P_2 C_{b,2} = P_2 Y_{b,2} - b^L$$

- Always constrained (β_b small)

Taylor rule:

$$1 + i_1^S = \frac{1}{\beta} \Pi_1^{\phi_\pi}, \qquad \phi_\pi > 1$$

- policy regime characterized by $\phi_\pi > 1$
- normalize $P_0 = 1$
- Assume $\Pi_2 = \Pi_1^\delta$; $\delta \geq 0$ (Krugman 1998 special case: $\delta = 0$)

Date-1 long bond price:

$$Q_1^L = \frac{1}{1 + i_1^S} = \beta \, \Pi_1^{-\phi_\pi}$$

$\Rightarrow$ Links monetary policy to liquidity risk: Q_1^L fluctuates with ξ_1 through Π_1
$\Rightarrow$ More aggressive targeting stabilizes the resale price of long bonds

Three pricing equations:

- **Short bond** (no supply):

$$1 = \mathbb{E}\left[(\lambda\, M^I_{01} + (1-\lambda)\, M^P_{01})\ \frac{1+i^S_0}{\Pi_1}\right]$$

- **Long bond** (hybrid: impatient sells at date 1, patient holds to date 2):

$$Q^L_0 = \mathbb{E}\left[\lambda\, M^I_{01}\, \frac{Q^L_1}{P_1}\ +\ (1-\lambda)\, M^P_{02}\, \frac{1}{P_2}\right]$$

- **Capital** (only patient types benefit, liquidity tax $(1-\lambda)$):

$$1 = (1-\lambda)\,\mathbb{E}\left[M^P_{02}\, R_K\right], \qquad R_K = \alpha \bar{A} K^{\alpha-1}$$

Equilibrium

allocations, prices, interest rates such that

1. households optimize,
2. bond markets clear, and
3. the Taylor rule holds

Zero-debt benchmark

Set $d_0 = 0$: no borrowing or lending. savers only save in capital.

from patient type's date-1 Euler + Taylor rule, solve for equilibrium inflation:

$$\Pi_1 = \bar{D}\,\xi_1^{a}, \qquad \text{where } a \equiv \frac{1-\rho}{\phi_\pi - \delta}, \quad \bar{D} \equiv \left(\frac{Y_2}{y_H}\right)^{\gamma/(\phi_\pi - \delta)}$$

inflation variance:

$$\mathrm{Var}(\log \Pi_1) = \left(\frac{1-\rho}{\phi_\pi - \delta}\right)^2 \mathrm{Var}(\log \xi_1) \quad \Longrightarrow \quad \text{strictly decreasing in } \phi_\pi$$

cumulative price level: $P_2 = \Pi_1 \Pi_2 = \Pi_1^{1+\delta}$

$$\mathrm{Var}(\log P_2) = (1+\delta)^2\, a^2\, \mathrm{Var}(\log \xi_1) \quad \text{amplified by persistence } \delta$$

Zero-debt: short-bond inflation risk premium

expected real return on the short nominal bond:

$$\bar{R}_0^{S*} = \bar{R}_1^f \cdot \underbrace{\frac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}}_{\mathrm{IRP}^S > 1}$$

- $\uparrow \phi_\pi \implies \downarrow \bar{R}_0^{S*}$ (lower compensation for inflation risk)
- as $\phi_\pi \to \infty$: $\mathrm{IRP}^S \to 1$. Perfect IT eliminates inflation risk premium
- IRP^S independent of γ, λ, or ω

Decomposing the real bond yield

use bond pricing equation $1 = \mathbb{E}[M \cdot (1 + i_0^S)/\Pi_1]$:

$$R_0^{S*} = \underbrace{\frac{1}{\mathbb{E}[M_{01}]}}_{R^f} + \underbrace{\left(-\frac{\mathrm{Cov}(M_{01},\ (1+i_0^S)/\Pi_1)}{\mathbb{E}[M_{01}]}\right)}_{\text{inflation risk premium } \psi}$$

at $b^L = 0$: $C_{s,1}$ is deterministic for each type $\implies$ $R^f = \bar{R}_1^f$ is invariant to ϕ_π

$\uparrow \phi_\pi \implies \downarrow \psi \implies \downarrow R_0^{S*}$

Why does ϕ_π lower the risk premium?

bonds as bad hedges:

- high ξ_1 = high marginal utility and high inflation
- $\implies$ nominal bonds pay poorly when consumption is most valuable
- $\implies$ positive inflation risk premium

but *which component* of the covariance responds to ϕ_π?

Channel	Depends on ϕ_π?	Contribution
Risk-free rate $\bar{R}_1^f$	No	0%
SDF volatility $\sigma_M \propto \sigma$	No	0%
Correlation $\mathrm{Corr}(M, \Pi_1^{-1}) = -1$	No	0%
Inflation vol $\sigma_{\Pi^{-1}} \propto a\,\sigma$	Yes	100%

safe asset demand story: $\uparrow \phi_\pi$ reduces volatility of real bond payoffs $\implies \bar{R}_0^{S*} \downarrow$
bonds remain bad hedges (correlation $= -1$)

Term structure of inflation risk premia

Short-bond IRP: $\mathrm{IRP}^S = \dfrac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}$

Long-bond IRP (patient, hold-to-maturity): $\mathrm{IRP}^L = \dfrac{\mathbb{E}[\xi_1^{-a(1+\delta)}]\,\mathbb{E}[\xi_1^{\rho}]}{\mathbb{E}[\xi_1^{\rho-a(1+\delta)}]}$

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Define $\Phi(\delta) \equiv \log \mathrm{IRP}^L - \log \mathrm{IRP}^S$:

(i) $\Phi(0) < 0$: long bond has *smaller* IRP

(ii) $\Phi'(\delta) > 0$: long-bond IRP strictly increasing in persistence

(iii) $\exists!\ \delta^*$: for $\delta > \delta^*$, standard ordering $\mathrm{IRP}^L > \mathrm{IRP}^S$

(iv) for $\delta > \delta^*$: $d\Phi/d\phi_\pi < 0$ — term premium compresses in ϕ_π

Zero-debt: capital is independent of ϕ_π

capital Euler (only patient types benefit; liquidity tax $1-\lambda$):

$$1 = (1-\lambda)\,\mathbb{E}\Big[M_{02}^{P}\,R_K\Big], \qquad R_K = \alpha \bar{A} K^{\alpha-1}$$

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at $b^L = 0$: $M^P_{02} = \beta^2 \xi_1^\rho (Y_2 + \bar{A}K^\alpha / C_{s,0})^{-\gamma}$

- R_K is independent of ϕ_π: capital has no inflation exposure
- R_K is increasing in λ: liquidity tax requires higher compensation
- R_K is independent of $\omega = \left(\frac{y_H}{y_L}\right)^\gamma \geq 1$: capital priced by patient type only

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Illiquidity premium: $R_K - \bar{R}^{L*}_0 > 0$

- R_K invariant to ϕ_π, while $\bar{R}^{L*}_0$ strictly decreasing
- $\Longrightarrow$ spread widens as inflation targeting becomes more aggressive
- R_K independent of ω, but bond price increasing in $\omega \Longrightarrow$ income risk amplifies

Taking stock: zero-debt results

At zero debt, more aggressive inflation targeting ($\uparrow \phi_\pi$):

1. $\bar{R}_0^{S*} \downarrow$: short-bond IRP falls (inflation volatility channel, 100%)
2. $\bar{R}_0^{L*} \downarrow$: long-bond IRP falls (same mechanism)
3. term premium compresses (when $\delta > \delta^*$)
4. R_K unchanged: capital return pinned by real fundamentals
5. illiquidity premium $R_K - \bar{R}_0^{L*}$ widens

Next: what happens when we turn on debt ($b^L > 0$)?

Positive debt: perturbation around stochastic steady state

expand around the stochastic zero-debt equilibrium (not a deterministic SS)

- inflation is stochastic even at $b^L = 0$
- $\implies$ nonzero inflation risk premium embedded from the outset
- contrasts with standard perturbation where risk premia vanish at first order

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at $b^L > 0$: impatient type's consumption becomes **stochastic**

$$C^I_{s,1} = y_L + \frac{Q^L_1}{P_1}\, b^L = y_L + \beta\, \Pi_1^{-(1+\phi_\pi)}\, b^L$$

- $C^I_{s,1}$ depends on Π_1 through liquidation payoff $\propto \Pi_1^{-(1+\phi_\pi)}$
- low base $y_L \implies$ marginal utility amplified by $\omega = (y_H/y_L)^\gamma$
- M^I_{01} now covaries with inflation $\implies$ interest rate risk channel activated

Two reinforcing channels

Proposition. At $b^L > 0$, $\partial \bar{R}_0^{L*} / \partial \phi_\pi < 0$ through two channels:

(i) Inflation risk channel (present for all λ):

- $\uparrow \phi_\pi$ reduces $\mathrm{Var}(\log \Pi_1) \implies$ compresses IRP
- same mechanism as zero-debt benchmark

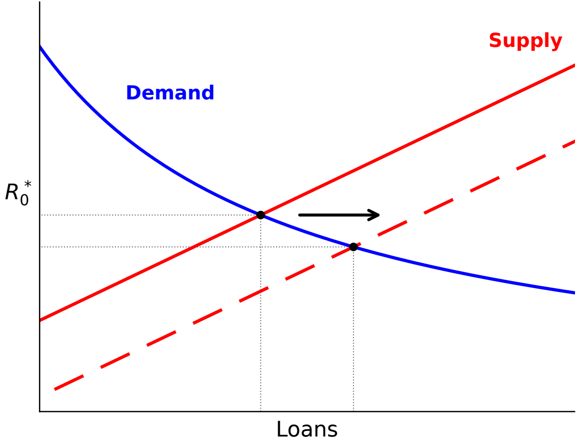
(ii) Interest rate risk channel ($\propto \lambda$, amplified by ω):

- $\uparrow \phi_\pi$ stabilizes $Q_1^L = \beta \Pi_1^{-\phi_\pi} \implies$ reduces capital-gain risk
- priced by impatient type's elevated SDF (ω times patient type's)
- vanishes at $\lambda = 0$; strengthens in λ and ω

Intuition: supply and demand for credit

more aggressive inflation targeting $\implies$ lower R_0^{L*}

$\implies$ more nominal debt supplied by savers $\implies$ lower R_0^L, more credit in equilibrium



[binding debt limit means the demand curve is $R_0^{L*} = d_0/\text{Loans}$]

Crowd-out: the question

when bonds become safer, do savers displace capital or just save more?

write total savings as $S_0 \equiv Y_0 - C_{s,0} = K + Q_0^L\, b^L$

differentiate with respect to bond holdings b:

$$\left.\frac{dK}{db}\right|_{b=0} = \underbrace{-1}_{\substack{\text{portfolio}\\ \text{margin}}} + \underbrace{\left.\frac{dS_0}{db}\right|_{b=0}}_{\substack{\text{savings}\\ \text{margin}}}$$

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- **portfolio margin** $= -1$: holding S_0 fixed, each dollar of bonds displaces a dollar of capital
- **savings margin** $= 1 + k$: when bonds are more attractive, saver may cut $C_{s,0}$ and save more
- $k \equiv dK/db$ is the **crowd-out coefficient**

Crowd-out: three forces

linearize capital Euler $(1-\lambda)\,\mathbb{E}[M^P_{02}\,R_K] = 1$ around $b = 0$ (stochastic):

$$0 = \underbrace{(1+k)\,Q^L_0 \cdot \frac{\gamma}{\bar{C}_0}}_{\text{(a) date-0 smoothing}} + \underbrace{k \cdot \frac{(1-\alpha)}{\bar{K}_1}}_{\text{(b) diminishing returns}} + \underbrace{k \cdot \frac{\gamma\,R_K}{\bar{C}_2}}_{\text{(c) date-2 smoothing}}$$

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(a) Date-0 consumption smoothing (drives crowd-out)

- saver saves more $(1+k>0) \implies$ cuts $C_{s,0} \implies$ higher marginal utility today
- resists further saving $\implies$ limits the savings margin

(b) Diminishing returns (absorbs displacement)

- capital falls $(k<0) \implies$ marginal product R_K rises $\implies$ partially stabilizes capital

(c) Date-2 consumption smoothing (limits crowd-out)

- capital falls $\implies$ $\bar{C}_2 = \bar{A}K^{\alpha} + Y_2$ falls $\implies$ higher marginal utility at date 2
- makes remaining capital more valuable $\implies$ resists further displacement

Capital extension: implications

$\uparrow \phi_\pi \implies$ safer nominal bonds $\implies b \uparrow \implies K \downarrow$ (one-for-one) $\implies \mathbb{E}_0[R_0^K] \uparrow$

mechanism: improved risk-return profile of nominal bonds crowds out capital investment

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Equilibrium implications ($\uparrow \phi_\pi$):

- $\bar{R}_0^{L*} \downarrow$ (declining real rate on nominal bonds)
- debt $\uparrow$ (rising private credit)
- $K \downarrow$ (investment drought)
- R_K stable (pinned by real fundamentals)
- $R_K - \bar{R}_0^{L*} \uparrow$ (widening safe–risky spread)
- term premium $\downarrow$ (when $\delta > \delta^*$)

Conclusion

aggressive inflation targeting regime:

- reduces inflation volatility $\implies$ lowers R_0^*
- low inflation vol & interest rate risk $\implies$ nominal bonds safer $\implies$ supply of savings in nominal assets goes up
- capital becomes relatively *more* risky $\implies$ supply of savings in real assets falls
- equilibrium: $\uparrow$ private credit, $\downarrow$ natural rate on nominal bonds, $\downarrow$ investment

a unified account of macro-finance trends since the 1980s:
declining real rates, rising debt, investment drought, stable returns on capital
all as equilibrium responses to a single shift in monetary policy regime

What next

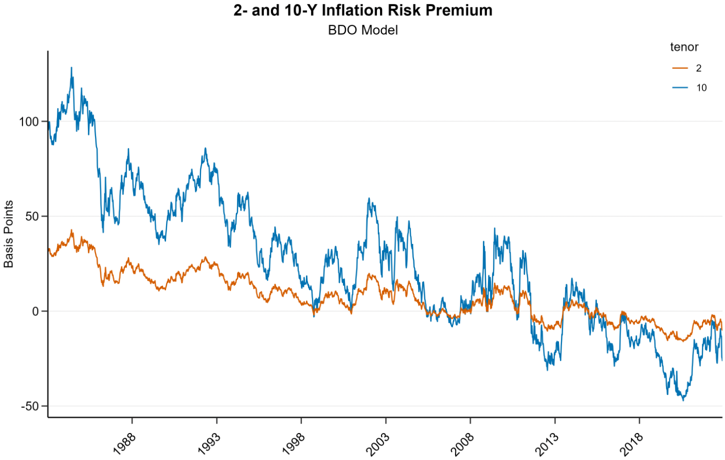
quantitative exercise:

- infinite horizon model with aggregate uncertainty
- decompose the contribution to decline in r^* from ϕ_π vs other forces (demography, productivity, capital risk)

THANK YOU!

Fact 1: Decline in long-term rates

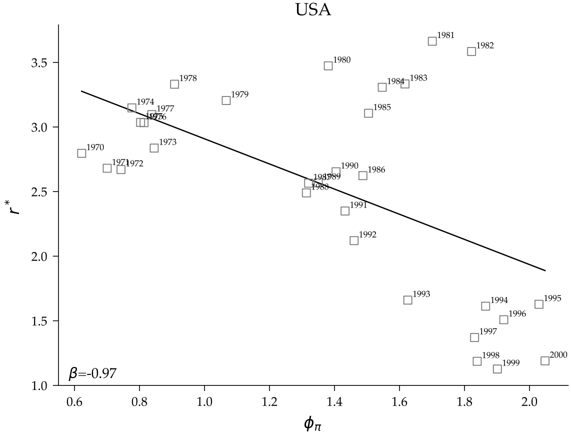
Inflation Risk Premium for the US

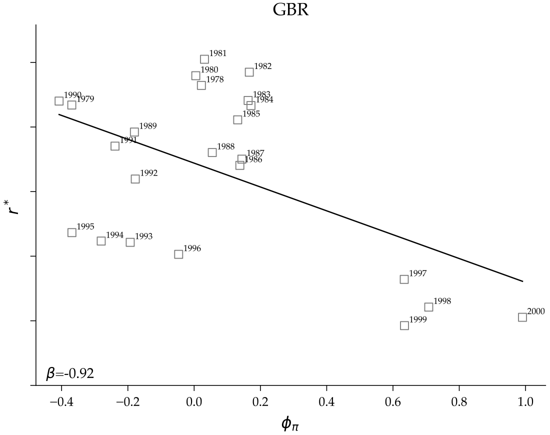


Breach, D'Amico, Orphanides (2020) for the US.

Monetary policy and r-star

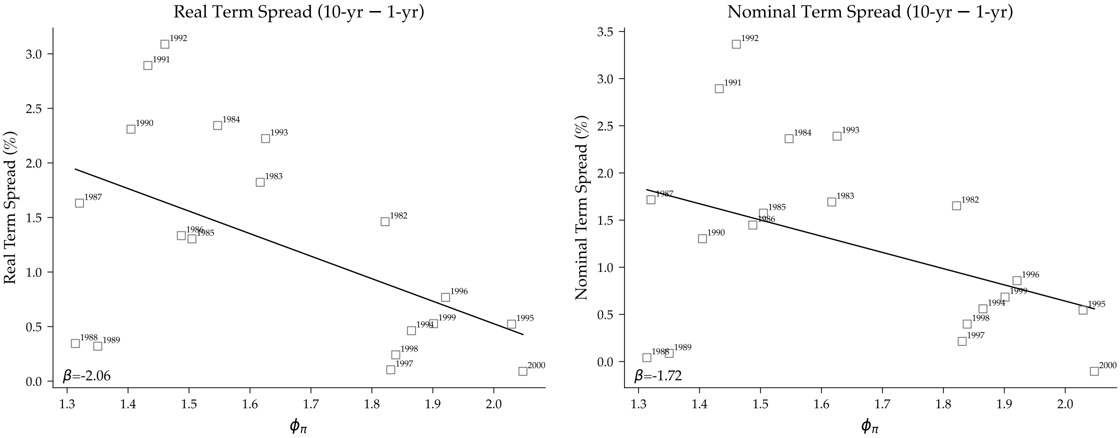
HLW estimates of r^* and our estimates of ϕ_π based on Carvalho Nechio Tristao (2021)





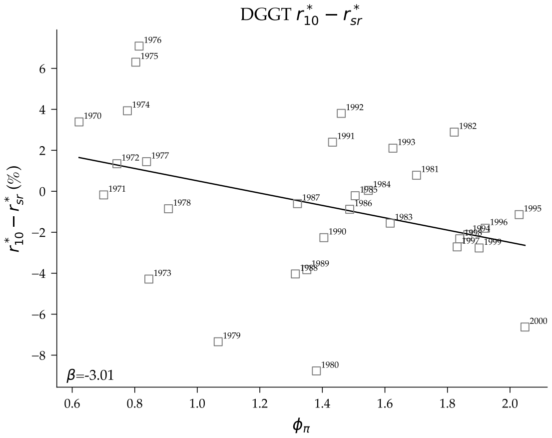
Monetary policy and term spread

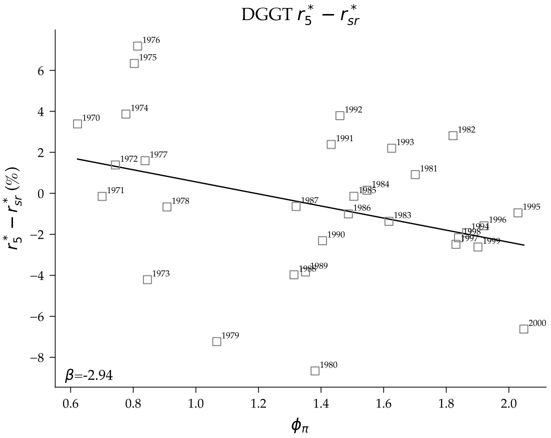
Term spread (10 y - 1 y) + our estimates of ϕ_π based on Carvalho Nechio Tristao (2021)



Monetary policy and DGGT r-star spread

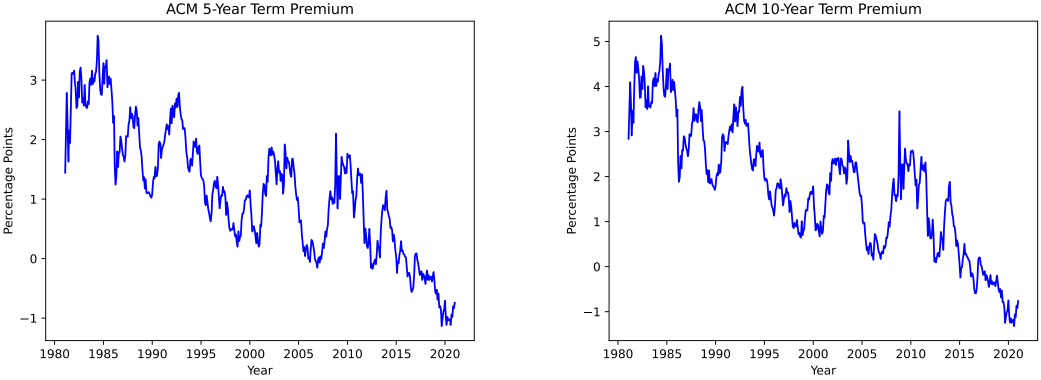
DGGT term spread + our estimates of ϕ_π based on Carvalho Nechio Tristao (2021)





Fact 1: Decline in long-term rates

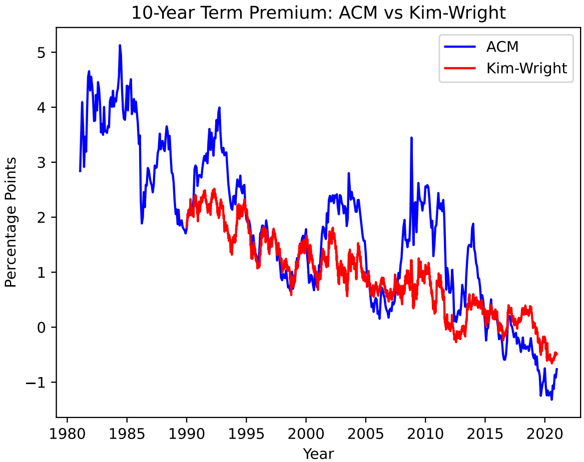
ACM term premium estimates on US nominal bonds



Term premium estimates from Adrian, Crump, and Moench (2013), Federal Reserve Bank of New York.

Appendix: ACM vs Kim-Wright Term Premium

10-year term premium: ACM and Kim-Wright estimates



ACM: Adrian, Crump, and Moench (2013). Kim-Wright: Kim and Wright (2005).

Zero-debt: long-bond yield

long-bond Euler:

$$Q_0^L = \mathbb{E}\left[\lambda\, M_{01}^I \, \frac{Q_1^L}{P_1} \;+\; (1-\lambda)\, M_{02}^P \, \frac{1}{P_2}\right]$$

at $b^L = 0$, substitute and collect powers of ξ_1:

	SDF	$\times$	Real payoff	$=$ ξ_1-exponent
Impatient	$M_{01}^I \propto \xi_1^{1}$	$\times$	$Q_1^L/P_1 = \beta\, \Pi_1^{-(1+\phi_\pi)} \propto \xi_1^{-a(1+\phi_\pi)}$	$\xi_1^{1-a-a\phi_\pi} = \xi_1^{\rho-a}$
Patient	$M_{02}^P \propto \xi_1^{\rho}$	$\times$	$1/P_2 = \Pi_1^{-(1+\delta)} \propto \xi_1^{-a(1+\delta)}$	$\xi_1^{\rho-a(1+\delta)}$

where $1 - a(1+\phi_\pi) = 1 - (1-\rho) - a = \rho - a$

Two cases:

- $\delta = 0$: both terms involve $\mathbb{E}[\xi_1^{\rho-a}]$
- $\delta > 0$: terms involve different moments — patient type exposed to $P_2 = \Pi_1^{1+\delta}$
- $\omega > 1$ $(y_L < y_H)$: impatient term has coefficient $\propto \omega$, patient term does not

Appendix: relation to alternative shock structures

baseline: $\xi_2 = \xi_1^{\rho}$ with $\rho < 1$ (mean reversion)

- high ξ_1 = bad times (high marginal utility) and high inflation
- bonds pay poorly in bad states $\implies$ positive risk premium
- $\uparrow \phi_\pi \implies \downarrow R_0^*$

alternative: $\xi_2 = \xi_1^{1+\rho}$ with $\rho > 0$ (persistence)

- high $\xi_1 \implies$ precautionary saving, deflationary pressure
- bonds are *good* hedges $\implies$ negative risk premium
- $\uparrow \phi_\pi \implies \uparrow R_0^*$ (opposite sign)

sign depends on whether bonds are bad hedges (baseline) or good hedges (alternative)

Zero-debt: pricing kernels

at $b^L = 0$: date-1 consumption differs only through idiosyncratic income

type-specific date-1 SDFs:

$$M_{01}^{I} = \beta\,\xi_1 \left(\frac{y_L}{C_{s,0}}\right)^{-\gamma}, \qquad M_{01}^{P} = \beta\,\xi_1 \left(\frac{y_H}{C_{s,0}}\right)^{-\gamma}$$

both $\propto \xi_1$ — differ only in ω:

$$M_{01}^{I} = \omega\, M_{01}^{P}, \qquad \omega \equiv \left(\frac{y_H}{y_L}\right)^{\gamma} \geq 1$$

Zero-debt: pricing kernels

at $b^L = 0$: date-1 consumption differs only through idiosyncratic income

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both $\propto \xi_1$ — differ only in ω:

$$M^I_{01} = \omega\, M^P_{01}, \qquad \omega \equiv \left(\frac{y_H}{y_L}\right)^{\gamma} \geq 1$$

average sdate 1 SDF :

$$M_{01} = \lambda\, M^I_{01} + (1-\lambda)\, M^P_{01} = \frac{\xi_1}{\bar{R}^f_1}; \quad \bar{R}^f_1 \equiv \frac{1}{\beta\,[\lambda\,\omega + (1-\lambda)]} \left(\frac{y_H}{C_{s,0}}\right)^{\gamma},$$

$\implies$ **at** $b^L = 0$: average SDF is proportional to ξ_1, regardless of λ or ω